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Repositionable windguard assembly for agricultural baler feeder system

Abstract

An agricultural machine includes: a machine frame; and a feeder system coupled with the machine frame and including a windguard assembly, the windguard assembly being configured for selectively occupying a first position and a second position by moving therebetween, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention pertains to an agricultural baler, and, more specifically, to a windguard assembly of the agricultural baler.

BACKGROUND OF THE INVENTION

(2) Agricultural harvesting machines, such as agricultural balers (which can be referred to as balers), have been used to consolidate and package crop material (which can also be referred to as forage, forage material, or forage crop material) so as to facilitate the storage and handling of the crop material for later use. Often, a mower-conditioner cuts and conditions the crop material and forms it into a swath or windrow to dry in the sun. When the cut crop material is properly dried (depending upon the application), an agricultural harvesting machine, such as an agricultural baler, travels along the swath or windrows (hereinafter, collectively referred to as windrows, unless otherwise specified) to pick up the crop material. Agricultural balers can be, for example, round balers, large square balers, and small square balers, which form the bales into corresponding shapes, as is known.

(3) More specifically, with respect to round balers, pickups of the baler gather the cut and windrowed crop material from the ground, and then convey the cut crop material toward a bale-forming chamber within the baler (that is, the bale chamber). A drive mechanism operates to activate any pickups of a pickup assembly, augers, and/or a rotor (which can be referred to as a feeder) of a rotary feeder assembly of a feeder system. A pickup can convey crop material in an

overshot manner, while a rotor can convey crop material toward or into the bale chamber in an undershot manner. A conventional bale chamber of a round baler may include a pair of opposing sidewalls with a series of rolls (which can be referred to as rollers) and belts that rotate and compress the crop material into a cylindrical shape. When the bale has reached a desired size and density, a wrapping assembly, which includes wrap material, may wrap the bale to ensure, at least in part, that the bale maintains its shape and density. The wrap material can include a film (such as a flexible plastic wrap) or a net (which can be referred to as net wrap). For example, wrap material may be used to wrap the bale of crop material. After wrapping, a cutting or severing mechanism of the wrapping assembly may be used to cut the wrap material once the bale has been wrapped. The wrapped bale may be ejected from the baler and onto the ground by, for example, raising a tailgate of the baler. The tailgate is then closed, and the cycle repeated as necessary and desired to manage the field of cut crop material.

(4) The feeder system can include not only the pickup assembly and the rotor assembly, for example, but also a floor, a cutting assembly, and a windguard assembly. The pickup includes a plurality of tines to pick up the crop material disposed in a windrow. The rotor, which is downstream of the pickup, can be positioned above the floor which the crop material traverses prior to entering the bale chamber and can work in conjunction with, and cooperatively with, the cutting assembly. The rotor can include a rotor shaft (extending transversely) and a plurality of tines (which can have a generally triangular or star-shaped configuration) spaced apart across the transverse extent of the baler. The tines are configured to engage and thereby to push the crop material towards the bale chamber and can be grouped in pairs, with a relatively short distance between the tines of a given pair. The floor can include a plurality of slots across the transverse extent of the floor, each slot extending longitudinally in the floor (that is, parallel or otherwise aligned with a direction of crop flow). The cutting assembly can include a plurality of knives (which can also be referred to as cutters) which are selectively received in the slots of this floor, respectively. When inserted through the slots so that the knives extend at least partially above the floor, each respective knife (depending upon the design) can extend between a pair of tines of the rotor, as the rotor shaft rotates the tines. Further, the knives, as they extend through the slots above the floor are configured to cut the crop material to a predetermined length, as the crop material passes by the knives prior to the crop material entering the bale chamber. These knives can be configured to drop at least partially below the floor upon being struck by a foreign object. The windguard assembly can include a roller and a plurality of tines extending aft and can be configured to retain the crop material along at least a part of its flow path as it proceeds rearward by the pickup and the rotor to the bale chamber and, in so doing, can be positioned adjacent to an upper portion of the crop material. Further, the windguard assembly can be pivotably attached to the pickup assembly.

(5) Occasionally, such as with round balers, the crop material becomes plugged in the feeder system. That is, the crop material bunches up so as to form a plug of crop material (which can be referred to as a crop plug), which does not allow, or otherwise at least partially inhibits, the crop material from flowing from the feeder system into the bale chamber. Round balers prone to plugging can have a variable position (drop) floor below the rotor (also known as the feeder) that allows the crop plug to be passed, by increasing a space between the floor and the rotor. Such a drop floor can be effective in unplugging if only the rotor is plugged but often is not effective for unplugging when the crop material is plugged between the windguard assembly and the pickup (resulting in a pickup plug). Such a plug can result, for example, when an operator's ground speed has exceeded the feeding capacity of the baler. When a plug of crop material occurs that cannot be cleared by the drop floor, some operators may use a manually or remotely operated rotor reverser of a reversing system (so as to reverse a direction of the flow of the crop material), which can be useful in loosening the crop material but typically is unable to clear a pickup entirely. The reversing system cannot effectively clear a pickup plug because a conventional construction of windguard

assembly geometry acts as a barrier, preventing the flow of the crop material in the reverse direction. For, the tines of the windguard assembly act as barbs, jamming the crop material and thus stopping the reverse flow of crop material and preventing the crop from clearing. Manual removal of the plug, thus, is often necessary for a pickup plug, which often requires the operator to remove the windguard assembly and to pull the crop material out of the pickup assembly by hand—a timely, labor-intensive job which often occurs in uncomfortable field conditions (heat, bugs, dust). (6) What is needed in the art is a way to reposition the windguard to facilitate removal of a plug of crop material from a feeder system when a drop floor is insufficient to do so, without having to clear the plug manually.

SUMMARY OF THE INVENTION

(7) The present invention provides an agricultural baler with a feeder system including a windguard assembly configured for selectively moving between a plurality of positions, such that a plug of crop material can be removed from the feeder system when the flow direction of the crop material in the feeder system is reversed.

(8) The invention in one form is directed to a windguard assembly of a feeder system of an agricultural baler, the agricultural baler including a baler frame, the feeder system being coupled with the baler frame, the windguard assembly including: the windguard assembly, which is configured for selectively occupying a first position and a second position by moving therebetween, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material.

(9) The invention in another form is directed to an agricultural machine, including: a machine frame; and a feeder system coupled with the machine frame and including a windguard assembly, the windguard assembly being configured for selectively occupying a first position and a second position by moving therebetween, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material.

(10) The invention in yet another form is directed to a method of using an agricultural baler, the method including the steps of: providing a baler frame and a feeder system, the feeder system being coupled with the baler frame and including a windguard assembly; occupying selectively, by the windguard assembly, a first position, the first position being associated with a normal flow direction of a crop material relative to the feeder system; and occupying selectively, by the windguard assembly, a second position by moving between the first position and the second position, the second position being associated with a reversal of the normal flow direction of the crop material.

(11) An advantage of the present invention is that it provides a windguard design of variable geometry for use with a harvester pickup to allow effective rejection of a crop plug from the machine.

(12) Another advantage is that it provides that, in varying the geometry of the windguard, the windguard, or at least a portion thereof, can be extended forward to open up space for the crop plug to be removed.

(13) Yet another advantage is that it provides that, in varying the geometry of the windguard, the tines of the windguard can be pivoted so that they do not act as barbs when the plug is attempted to be removed.

(14) Yet another advantage is that the windguard assembly does not need to be removed in order to remove a crop plug.

(15) Yet another advantage is that a variation of the geometry of the windguard assembly can be controlled by a controller, based upon sensor feedback, in order to optimize crop feeding.

Alternatively, the variation of the geometry of the windguard assembly can be controlled by a user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For the purpose of illustration, there are shown in the drawings certain embodiments of the present invention. It should be understood, however, that the invention is not limited to the precise arrangements, dimensions, and instruments shown. Like numerals indicate like elements throughout the drawings. In the drawings:

(2) FIG. 1 illustrates a schematic side view of an exemplary embodiment of an agricultural vehicle, formed as a tractor, and an agricultural baler, the agricultural baler including a feeder system including a windguard assembly, in accordance with an exemplary embodiment of the present invention;

(3) FIG. 2 illustrates a perspective view of an exemplary embodiment of the feeder system of the agricultural baler, the feeder system including the windguard assembly in a first position, with portions broken away, in accordance with an exemplary embodiment of the present invention;

(4) FIG. 3 illustrates a perspective view of the feeder system of FIG. 2, the windguard assembly being in a second position, with portions broken away, in accordance with an exemplary embodiment of the present invention;

(5) FIG. 4 illustrates a side view of the feeder system of FIG. 2, the windguard assembly being in the first position, with portions broken away;

(6) FIG. 5 illustrates a side, partially exploded, view of the feeder system of FIG. 2, the windguard assembly being in the second position, with portions broken away; and

(7) FIG. 6 illustrates a flow diagram showing a method of using an agricultural baler, in accordance with an exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(8) The terms “forward”, “rearward”, “left” and “right”, when used in connection with the agricultural vehicle, agricultural baler, and/or components thereof are usually determined with reference to the direction of forward operative travel of the agricultural vehicle and/or agricultural baler, but they should not be construed as limiting. The terms “longitudinal” and “transverse” are determined with reference to the fore-and-aft direction of the agricultural vehicle and/or agricultural baler and are equally not to be construed as limiting. The terms “downstream” and “upstream” are determined with reference to the intended direction of crop material flow during operation, with “downstream” being analogous to “rearward” and “upstream” being analogous to “forward.”

(9) Referring now to the drawings, and more particularly to FIG. 1, there is shown an embodiment of an agricultural vehicle **100** (which can be referred to as a work vehicle **100**) towing an agricultural baler **101**, in accordance with the present invention, to perform a baling operation within a field (FIG. 1 is generic to all embodiments of the present invention discussed herein). As shown, work vehicle **100** can be configured as an agricultural tractor, such as an operator-driven tractor or an autonomous tractor. However, in some embodiments, the work vehicle **100** may correspond to any other suitable vehicle configured to tow a baler across a field or that is otherwise configured to facilitate the performance of a baling operation, including an autonomous baling vehicle. Additionally, as shown, baler **101** can be configured as a round baler configured to generate round bales. However, in some embodiments, baler **101** may have any other suitable configuration, including being configured to generate square or rectangular bales. It should be further appreciated that baler **101**, while shown as being towed by tractor **100**, may also be a self-propelled baler that does not rely on a separate vehicle for propulsion and/or power to function. While the present invention is described with respect to a baler, it can be appreciated that other the present invention can be used in conjunction with other agricultural machines, such as vehicles and/or implements, such as a self-loading forage wagon.

(10) Work vehicle **100** includes a pair of front wheels **102**, a pair of rear wheels **103**, and a chassis **104** coupled to and supported by the wheels **102**, **103**. An operator's cab **105** may be supported by a portion of the chassis **104** and may house various input devices for permitting an operator to control the operation of work vehicle **100** and/or baler **101**. Additionally, work vehicle **100** may include an engine and a transmission mounted on chassis **104**. The transmission may be operably coupled to the engine and may provide variably adjusted gear ratios for transferring engine power to wheels **103** via a drive axle assembly.

(11) As shown in FIG. 1, work vehicle **100** may be coupled to baler **101** via a power take-off (PTO) **106** and a tongue **107** to a hitch of work vehicle **100** to allow vehicle **100** to tow baler **101** across the field. As such, work vehicle **100** may, for example, guide baler **101** toward crop material **136** deposited in windrows on the field. As is generally understood, to collect the crop material **136**, baler **101** includes a feeder system **108** (which can be referred to generally as a crop collector) mounted on a front end of baler **101**. Feeder system **108** may, for example, include a pickup assembly **111**, a rotary feeder assembly **121**, a floor **131**, and a windguard assembly **128**. Pickup assembly **111** includes a rotating wheel with tines (**240**) that collect crop material **136** from the ground and direct crop material **136** toward a bale chamber **109** of baler **101** in an overshot manner (rotating clockwise in FIG. 1). In FIG. 1, the tines of pickup assembly **111** rotate in a clockwise direction (as indicated by flow direction **132** of crop material **136**), to facilitate crop material **136** to move in flow direction **132**. These tines, however, can be reversed (such as by way of operator, control system **122** of baler **101** and/or control system **114** of tractor **100**), such that the tines rotate in a counter-clockwise direction in FIG. 1, which can facilitate a reversed flow direction **133** of crop material **136**. Rotary feeder assembly **121** includes a rotating shaft (rotor shaft) and a plurality of rotor devices **144** (which can be referred to as rotors **144**) mounted to the shaft so as to rotate about the shaft in a counter-clockwise manner (as seen in FIG. 1 and shown by arrow **134**), rotors **144** configured to push or otherwise to move crop material **136** towards or into bale chamber **109**, in an undershot manner (rotating counter-clockwise in FIG. 1). Feeder system **108** can also include a rotating shaft (not shown in FIG. 1) generally between pickup assembly **111** and rotary feeder assembly **121** that includes side augers for moving crop material **136** inwardly prior to entering bale chamber **109**. FIG. 1 shows crop material **136** lying in a windrow on the ground of the field and being picked up by pickup assembly **111** of feeder system **108**. Upon being picked up, crop material flow **136** flows over pickup assembly **111**, under windguard assembly **128**, and under rotary feeder assembly **121**, as indicated by arrow **132**. Arrow **132** thus indicates the normal flow direction **132** of crop material **136** relative to feeder system **108**. Conversely, arrow **133** shows the flow direction which is opposite the normal flow direction of crop material **136**; this opposite flow direction **133** can also be referred to as a reverse flow direction **133** (that is, a reversal of normal flow direction **132**). Thus, the flow direction (whether normal **132** or reversed **133**) of crop material **136** can be facilitated by the normal or opposition rotational direction of pickup assembly **111** and/or rotary feeder assembly **121**. FIG. 1 shows baler **101** schematically and thus with one pair of wheels, though it can be appreciated that baler **101** can include one or more additional pair of wheels, as in FIG. 2.

(12) Inside bale chamber **109**, rollers, belts, and/or other devices compact the crop material **136** to form a generally cylindrically-shaped bale **110**. Bale **110** is contained within baler **101** until ejection of bale **110** is instructed (e.g., by the operator and/or a baler controller **123** of baler **101**). In some embodiments, bale **110** may be automatically ejected from baler **101** once bale **110** is formed, by baler controller **123** detecting that bale **110** is fully formed and outputting an appropriate ejection signal. Further, work vehicle **100** includes a control system **114**, which includes a controller **115**, which includes a processor **116**, memory **117**, data **118**, and instructions **119**. Control system **114** can further include an input/output device **120** such as a laptop computer (with keyboard and display) or a touchpad (including keypad functionality and a display), device **120** being configured for a user to interface therewith.

(13) As shown in FIG. 1, baler **101** may also include a tailgate **112** movable between a closed position (as shown in the illustrated embodiment) and an opened position via a suitable actuator assembly. Tailgate **112** and/or the actuator assembly may be controlled to open and close by baler controller **123**. In the closed position, tailgate **112** may confine or retain bale **110** within baler **101**. In the open position, tailgate **112** may rotate out of the way to allow bale **110** to be ejected from the bale chamber **109**. Additionally, as shown in FIG. 1, baler **101** may include a ramp **113** extending from its aft end that is configured to receive and direct bale **110** away from baler **113** as it is being ejected from bale chamber **109**. In some embodiments, ramp **113** may be spring loaded, such that ramp **113** is urged into a raised position, as illustrated. In such embodiments, the weight of bale **110** on ramp **113** may drive ramp **113** to a lowered position in which ramp **113** directs bale **110** to the soil surface. Once bale **110** is ejected, bale **110** may roll down ramp **113** and be deposited onto the field. As such, ramp **113** may enable bale **110** to maintain its shape and desired density by gently guiding bale **110** onto the field. Further, baler **101** includes a control system **122**, which includes controller **123**, which includes a processor **124**, memory **125**, data **126**, and instructions **127**. Controller **123** can communicate with controller **115**, so that controller **115** outputs information to the display of input/output device **120** of work vehicle **100**, thereby informing a user of various conditions of baler **101** and bales **110** forming or formed therein. Further, baler **101** includes a frame **130** (which can be referred to as baler frame **130**, or more generally as a machine frame **130**) to which all of the components of baler **101** are directly or indirectly coupled. Thus, feeder system **108** and thus also pickup assembly **111**, rotary feeder assembly **121**, and windguard assembly **128** are coupled with frame **130**.

(14) It should be appreciated that the configuration of work vehicle **100** described above and shown in FIG. 1 is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of work vehicle configuration. For example, in an alternative embodiment, a separate frame or chassis may be provided to which the engine, transmission, and drive axle assembly are coupled, a configuration common in smaller tractors. Still other configurations may use an articulated chassis to steer work vehicle, or rely on tracks in lieu of wheels **102**, **103**. Additionally, as indicated previously, work vehicle **100** may, in some embodiments, be configured as an autonomous vehicle. In such embodiments, work vehicle **100** may include suitable components for providing autonomous vehicle operation and, depending on the vehicle configuration, need not include the operator's cab **105**.

(15) Additionally, it should be appreciated that the configuration of baler **101** described above and shown in FIG. 1 is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of baler configuration, or other agricultural machines, such as a vehicle and/or implement, such as a self-loading forage wagon. For example, as indicated previously, baler **101** may, in some embodiments, correspond to a square baler configured to generate square or rectangular bales. It should be further appreciated that the illustration of baler **101** in FIG. 1 is schematic.

(16) Further, in general, controllers **115**, **123** may each correspond to any suitable processor-based device(s), such as a computing device or any combination of computing devices. Each controller **115**, **123** may generally include one or more processor(s) **116**, **124** and associated memory **117**, **125** configured to perform a variety of computer-implemented functions (e.g., performing the methods, steps, algorithms, calculations and the like disclosed herein). Thus, each controller **115**, **123** may include a respective processor **116**, **124** therein, as well as associated memory **117**, **125**, data **118**, **126**, and instructions **119**, **127**, each forming at least part of the respective controller **115**, **123**. As used herein, the term “processor” refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, the respective memory **117**, **125** may generally include memory element(s) including, but not limited to, computer readable medium (e.g., random access

memory (RAM)), computer readable non-volatile medium (e.g., a flash memory), a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD), and/or other suitable memory elements. Such memory **117, 125** may generally be configured to store information accessible to the processor(s) **116, 124**, including data **118, 126** that can be retrieved, manipulated, created, and/or stored by the processor(s) **116, 124** and the instructions **119, 127** that can be executed by the processor(s) **116, 124**. In some embodiments, data **118, 126** may be stored in one or more databases.

(17) Baler controller **123**, herein, is assumed to be the primary controller for controlling operations of baler **101**. However, controller **123** of baler **101**, as indicated in FIG. 1, can be in communication with controller **115** of work vehicle **100**, thereby forming a unified control system **129**, such that any or all information associated with either controller **115, 123** can be shared with the other controller **115, 123**, and either controller **115, 123** can perform the functions of the other controller **115, 123**. Controllers **115, 123** can communicate with each other in any suitable manner, such as a wired connection or a wireless connection, such as radio signals (RF), light signals, cellular, WiFi, Bluetooth, Internet, via cloud-based devices such as servers, and/or the like. Further, while not shown, both controllers **115, 123** can communicate with a remotely located data center (which can also be a part of unified control system **129**), which controllers **115, 123** can communicate with by any suitable way, such as those just referenced. Such a data center can include its own controller (and thus processor(s), memory, data, and instructions, substantially similar to that described above with respect to controllers **115, 123**) which can be configured to perform any of the functions associated with controllers **115, 123**. Controllers **115, 123** and the data center can be a part of any network facilitating such communication therebetween, such as a local area network, a metropolitan area network, a wide area network, a neural network, whether wired or wireless.

(18) Referring now to FIG. 2, there is shown baler **101** with portions broken away. Baler **101** includes baler control system **122** and feeder system **108**. Control system **122**, which includes controller **123** and can include sensor(s) **535A, 535B** (FIG. 5), can be a part of unified control system **1129**, which also includes tractor control system **114**. Feeder system **108** includes pickup assembly **111**, rotary feeder assembly **121** (FIG. 1), and windguard assembly **128**, which is shown in first position **237**. Pickup assembly **111** includes a plurality of pickup tines **240**, pickup frame **241**, and a latch **271**. Pickup tines **240** are configured to pick up the windrow of crop material **136** from the ground as tines **240** rotate in a counter-clockwise direction (as viewed from the bottom left of the page of FIG. 2), with crop **136** moving in a normal flow direction **132** once picked up by tines **240**. Pickup frame **241** (which can be referred to as frame **241**) includes a windguard mounting device **242** on each lateral side of feeding system **108**, which can be fastened to other portions of pickup frame **241**, such as by way of rivets or threaded fasteners, such as bolts or screws, or the like. Windguard mounting device **242** (as indicated in FIG. 2) can include two plates spaced transversely apart and coupled with one another, such as by way of joiner devices positioned at the bottom left and bottom right of windguard mounting device **242** in FIG. 2, as well as a stop mechanism **243**. Windguard mounting device **242** further includes stop mechanism **243** configured to selectively receive thereon a portion of windguard assembly **128**; windguard mounting device **242** can be attached to pickup frame **241** in any suitable manner, such as by way of welding or by way of fasteners. Pickup frame **241** and windguard mounting device **242** can each be made of any suitable material, such as steel or a polymer. Latch **271** is configured for selectively latching (that is, holding in place) shaft **275** of arm **270** in first position **237** and thus in rearward position **272**, as indicated in FIG. 2. Latch **271** is pivotably coupled with windguard mounting device **242**, such as by way of shaft, so as to rotate about an axis of rotation **279** extending transversely through this shaft. Latch **271**, as indicated in FIG. 2, can include two like plates spaced transversely apart and positioned within and respectively adjacent to the two plates of windguard mounting device **242**. The two plates of latch **271** can be rigidly coupled to one another by the shaft forming axis of rotation **279**, this shaft being mounted in respective mounting holes formed by each

of the plates of latch **271**. In FIG. 2, latch **271** is in its normal position, a latching position, and thus pivoted downwards. In this way latch **271** can be biased into this latching position in any suitable manner, such as by way of a spring (not shown). Upon applying a force, latch **271** can be pivoted upwardly (counter-clockwise, viewed from the bottom left in the page of FIG. 2) so as to release shaft **275**, so that shaft **275** can slide forward. Latch **271** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding, and/or injection molding. Further, though not shown in FIG. 2, rotary feeder assembly **121** of feeder system **108** includes rotors **144**, each of which includes a plurality of generally triangular shaped tines (as is known), the rotors **144** being configured to move crop material **136** towards bale chamber **109** as rotors **144** rotate. Thus, for example, rotors **144** move crop material **136** toward bale chamber **109** in normal operation, and when reversed rotors **144** can facilitate removal of a crop plug.

(19) Windguard assembly **128** includes an arm **270**, roller **247**, tine mechanism **248**, a chain assembly, linkage assembly **250**, actuator **251** (according to an optional embodiment), and connections A, B, C, D, E, F, G. Windguard assembly **128** can be substantially similar on both lateral sides of windguard assembly **128**, such that a description of one lateral side of windguard assembly **128** substantially serves as a description of the opposing lateral side of windguard assembly **128**. Windguard assembly **128** is configured for selectively occupying a first position **237** and a second position **338**, the first position **237** being associated with normal flow direction **132** of a crop material **136** relative to feeder system **108**, second position **338** being associated with a reversal of the normal flow direction **132** of the crop material **136** (and thus a reversed flow direction **133**). In general, first position **237** is when windguard assembly **128** occupies, at least in part, a rearward position **272** (as in FIG. 2), and second position **338** is when windguard assembly **128** occupies, at least in part, a forward position **373** (as in FIG. 3).

(20) Arm **270** can be positioned between the two plates of windguard mounting device **242**. Arm includes a first section **245** and a second section **246** which is disposed at an angle (not 180 degrees) relative to first section **245** and can be formed integral with first section **245** (such that first and second sections **245**, **246** are not pivotable relative to one another). Arm **270** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding, and/or injection molding. First section **245** can be positioned closer to, or adjacent to, an inboard one of the plates of windguard mounting device **242**. First section **245** can include a transversely extending pin, shaft, or the like (hereinafter, shaft **275**), by way of which first section **245** is pivotably and slidably coupled with windguard mounting device **242** at a slot **274** formed respectively by each of the plates of windguard mounting device **242**, the opposing ends of shaft **275** extending respectively into slot **274** formed by these plates of windguard mounting device **242**; this coupling between shaft **275** of arm **270** and slot **274** is connection **265A**. Thus, arm **270** is configured for moving, such as sliding, between a rearward position **272** and a forward position **373** by way of shaft **275** and slot **274**. That is, arm **270**, by way of a shaft **275** of first section **245**, pivots and slides within slot **274** formed in windguard mounting device **242**. When arm **270** is in rearward position **272**, shaft **275** is positioned at a rearward end of slot **274**; and, when arm **270** is in forward position **373**, shaft **275** is positioned at a forward end of slot **274**. Second section **246**, in FIG. 2, is shown to be resting on stop mechanism **243** (if second section **246** were not resting on stop mechanism **243**, then second section **246** can be supported and thus suspended by a chain **461** (FIG. 4) or can be supported by crop material **136**). Second section **246** is pivotably connected with roller **247** at connection **265B**. Connection **265B** can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. Roller **247** extends between opposing second sections **246** across a front portion of windguard assembly **128** and can pivot in either direction, as is known, as crop material **136** contacts roller **247**. Though the windguard assembly of the present invention is discussed as having a roller, it can be appreciated that the present invention is not limited to a type of windguard

assembly having a roller (and thus includes windguard assemblies without a roller). With arm **270** in rearward position **272** in FIG. 2, tine mechanism **248** is in down position **267**. Thus, arm **270** is configured for being pivotably coupled with pickup assembly **111** when latched by latch **271** (by way of shaft **275** in slots **274**) and is configured for being slidably coupled with pickup assembly **111** (by way of shaft **275** in slots **274**) when unlatched by latch **271**.

(21) Tine mechanism **248** includes a bar **259** (which can also be referred to as a tube **259**, or tine bar **259**) and a plurality of tines **260** rigidly connected to bar **259**, tines **260** projecting generally in an aft direction. Bar **259** is pivotably coupled with second section **246** at connection **265C**, such that bar **259** can extend through each second section **246** (this is different than a known prior art device, wherein a tine bar is welded to an arm). Connection **265C** can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. Tines **260** are configured for riding on a top portion of crop material **136** as crop material flows through feeder system **108** and thereby serves, at least in part, to retain crop material **136** within feeder system **108** and on a proper path to bale chamber **109**. Tine mechanism **248** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding, and/or injection molding. Tine mechanism **248** can further include a hook or loop **286** configured to receive therethrough a link of a bottom end of a chain, such as chain **461** (FIG. 4) of a chain assembly. Tine mechanism **246** can include two such loops **286** on opposing sides of tine mechanism **248** (as seen in FIG. 2), each loop **286** receiving therethrough a bottom end of a respective chain **461**.

(22) The chain assembly can include one or more chains **461** (FIG. 4), such as two chains, as indicated by two loops **286** shown in FIG. 2). Each chain **461** can extend between a respective loop **286** of tine mechanism **248** and a front portion of baler **101** on baler frame **130** so as to be pivotal at each such connection. Whereas the bottom end of chain **461** connects with loop **286**, the top end of chain **461** connects with a hook **487** connected to the front portion of baler **101** on baler frame **130**. Hook **487** can be connected to a bracket with a flat surface facing forward, and this bracket can be connected with a transversely extending bracket of baler **101**. Chain **461** can thereby hold (suspend) and thus support at least portions of windguard assembly **128** when such portions of windguard assembly **128** are not otherwise supported, for example, by stop mechanism **243**, crop material **136**, and/or (if provided) actuator **251**. When chain **461** supports portions of windguard assembly **128**, chain **461** can be taut, and when it does not chain **261** can be slack. The chain assembly can also include a chain sleeve surrounding a portion of chain **461**, such as near a bottom end of chain **461**; this chain sleeve can be a relatively rigid material, such as a plastic pipe, and or made of flexible material. For high volume plugs of crop material **136**, windguard roller **247** may roll over the top of the plug, in which case support chains **461** can be slack.

(23) Linkage assembly **250** includes a first linkage **276**, a second linkage **277**, a third linkage **278**, and a fourth linkage **264**. Linkage assembly **250** can be positioned to the inside (inboard side) of arm **270**, as shown with linkage assembly **250** in the foreground of FIG. 2. Alternatively, linkage assembly **250** can be positioned entirely to the outside (outboard side) of arm **270** (not shown). First linkage **276** is pivotably coupled at a proximal end of first linkage **276**, at a pivot connection, with pickup assembly **111** by way of a tube **288** (which can have a hemispherical cross-section as shown, so that shaft **275** can slide forward in slot **274**) of pickup assembly **111**. Tube **288** can extend through both plates of windguard mounting device **242**, can be attached to the exterior of each plate (not running through the interior), or can be attached only to the exterior of the inboard plate. First linkage **276**, at its proximal end, can have a hole through which tube **288** extends, so that first linkage is pivotably coupled with tube **288** (the proximal end of first linkage **276** can have a widened end so as to accommodate the radius of tube **288**). Thus, first linkage **276** can be positioned to the inboard side of the inboard plate of windguard mounting device **276** and pivotably mount to tube **288** (alternatively, rather than being coupled with tube **288**, first linkage **276** can be coupled with a bracket, such as the bracket shown in FIG. 2 projecting to the rear of windguard

mounting device **242**). Further, first linkage **276** is pivotably coupled with second linkage **277** at pivot connection **265D**. Second linkage **277** is pivotably coupled with arm **270** at pivot connection **265E** and with third linkage **278** at pivot connection **265F**. Third linkage **278** is pivotably coupled with fourth linkage **264** at pivot connection **265G**. Fourth linkage **264** is rigidly coupled with tine bar **259**, such as by welding. Linkage assembly **250** is configured for moving tine mechanism **248** between down position **267** and up position **368** when arm **270** slides between rearward position **272** and forward position **373**. Thus, the present invention provides linkage assembly **250** which is configured for rotating and thereby for changing an angle of tines **260** dependent on pivot connection **265C**. First, second, third, and fourth linkages **276**, **277**, **278**, **264** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding and/or injection molding.

(24) Actuator **251** is shown schematically in FIG. **2**. Actuator **251** can be in any form (such as a linear or a rotary actuator, a hydraulic actuator, a pneumatic actuator, or an electric actuator), but is described herein as a hydraulic actuator assembly **251**, with a cylinder, a piston in the cylinder, and a rod attached to and movable with the piston, as is known. Actuator **251** can be mounted to the outside or inside of pickup frame **241**, optionally in whole or in part to windguard mounting device **242** (only one actuator **251** can be used, or an actuator **251** can be positioned at each transverse end of windguard assembly **128**). The cylinder can be positioned aft, and the rod can be positioned fore; thus, a distal end of the rod can be coupled (for example, pivotably) with shaft **275** so as to be able to move arm **270** forwardly and rearwardly as the rod extends and retracts relative to the cylinder. In this way a rearward end of the cylinder of hydraulic cylinder assembly **251** can be coupled with frame **241** (or, more specifically, windguard mounting device **242**), and a forward end of the rod can be coupled (i.e., pivotably) with shaft **275**, or any other portion of first section **246** of arm **270**. Nevertheless, using an actuator, such as actuator **251**, is but one embodiment of the present invention. Thus, actuator **251**, according to another embodiment of the present invention, can be omitted altogether. Omitting actuator **251**, windguard assembly **128** can be slid forward and backward manually by an operator, according to one embodiment of the present invention.

(25) On the other hand, according to another embodiment of the present invention that omits actuator **251**, windguard assembly **128** can be slid forward and backward (the repositioning of windguard assembly **128**) by the movement of crop material **136**, rather than by an actuator or manually. Thus, windguard assembly **128** can be thought of as floating on top of crop material **136**. Once latch **271** releases windguard assembly **128** and as the crop flow direction is reversed (**133**), windguard assembly **128** will float on top of the reversing crop material **136**, and the movement of crop material **136** can cause windguard assembly **128** to be moved forward (from rearward position **272** to forward position **373** and thus from first position **237** to second position **338**, which also entails tine mechanism **248** moving from down position **267** to up position **368**). This movement forward of windguard assembly **128** can be further enhanced (assisted) if the travel direction of baler **101** is reversed (by reversing tractor **100**). Thus, the pile (plug) of crop material **136** will assist with moving windguard assembly **128** out of the way from plugging feeder system **108**. In order to move windguard assembly **128** back to its normal operating position (that is, first position **237**, which entails windguard assembly being in rearward position **272** and tine mechanism **248** being in down position **267**), tractor **100** can be driven forward (causing baler **101** to move forward) into the ejected crop pile/windrow, windguard assembly **128** will be pushed (by way of crop material **136**) back up through slots **274** and will be latched by latch **271** and into the normal operation position. Thus, in operation, when a plug is detected (manually by the operator, or by a sensor (not shown)), a controller (such as baler controller **123** and/or tractor controller **115**) can be used to reverse the normal rotor direction **134** of rotors **144** of rotary feeder assembly **121** (by way of an actuator (not shown)) and/or the normal direction of tines **240** of pickup assembly **111** (by way of an actuator (not shown)) in order to reverse the flow direction (**133**) of crop material **136** without an actuator for directly moving windguard assembly **128** in an unplugging event. In this

way, the sliding forward/backward of windguard assembly **128** can be accomplished by interaction with the flow direction of crop material **136**, and the operator can control the unplugging while remaining in the comfort of cab **105**.

(26) Referring now to FIG. **3**, there is shown baler **101** with feeder system **108**, with windguard assembly **128** being in second position **338**. As such, arm **270** is in forward position **373**, and tine mechanism **248** is in up position **368**. Roller **247** (and thus also, in part, arm **270**) can be supported by crop material **136** (such as a crop plug), or second section **246** of arm **270** can be supported, in part, by a chain **461** (FIG. **4**). In FIG. **3**, arm **270** at shaft **275** has slid forward in slots **274**. With the use of latch **271**, latch **271** has to release shaft **275** in order for arm **270** to slide forward in slots **274**. Thus, latch **271** must be rotated upwardly (counter-clockwise), and then arm **270** can be slid forward. Once arm **270** has slid forward, latch **271** can be released to return to its down position (as shown in FIG. **3**). Moving latch **271** upward in order to release arm **270** can be accomplished in any suitable way; for example, (1) a user can pull on a rope, cable, or the like, by hand to pull latch **271** upwardly, and this rope or cable can be attached to latch **271** at hole **381**; (2) an actuator (not shown) can move latch **271** in order to release arm **270**, the actuator being of any suitable type (i.e., hydraulic, pneumatic, electric, linear, and/or rotary); or (3) latch **271** can be moved by being connected to and thus triggered by another baler function. As arm **270** slides forward in slots **274**, first linkage **276** begins a chain of causation through second, third, and fourth linkages **277**, **278**, **264** to cause tine bar **259** to rotate upwardly to up position **368**. Further, an inside of each arm **270** can include thereon a stop device **385** (shown in broken lines), which can help linkage assembly **250** from overextending in either direction (by contacting first linkage **276** in first position **237**, and/or by contacting second linkage **277** in second position **338**). Stop device **385** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be formed integral with arm **270** or welded thereto, for example.

(27) Referring now to FIG. **4**, there is shown a side view of baler **101**, in particular, feeder system **108**, with portions broken away. Windguard assembly **128** is shown in two orientations, both of which are in first position **237** (which is the normal operation position for windguard assembly **128**). Regarding the first orientation (the side view of what is shown in FIG. **2**, with windguard assembly **128** shown in solid lines), second section **246** of arm **270** is resting on stop mechanism **243**. This orientation can occur when windguard assembly **128** encounters little or no crop material **136**. Regarding the second orientation, arm **270** is pivoted further upwardly (in a counter-clockwise direction) about shaft **275** and connection **265A**, which can be to a maximum extent. This orientation can occur when windguard assembly **128** encounters a relatively large volume of crop material **136**, which can cause arm **270** to pivot upward. Further, though arrow **132** is not shown in FIG. **4**, crop material **136** is understood to flow in the normal flow direction **132** relative to pickup assembly **111**, rotors **144**, and floor **131**. Further, latch **271** can pivot about axis of rotation **279**.

(28) Referring now to FIG. **5**, there is shown a side view of baler **101**, in particular, feeder system **108**, with portions broken away. Windguard assembly **128** is shown in second position **338**, with arm **270** being slid to forward position **373**, and thus with tine mechanism **248** in up position **368** (this side view is substantially similar to what is shown in FIG. **4**). Latch **271** is shown exploded from windguard mounting device **242**. Latch **271** is shown to include hole **381** for a rope or cable, a hole **582** in which to make the pivot connection between latch **271** and windguard mounting device **242**. Latch **271** also includes a mouth **584** which can capture arm **270**, such as by way of shaft **275**.

(29) Further, FIG. **5** shows that baler control system **122** can further include sensor(s) **535A** and/or **535B** (shown schematically) which are operatively coupled with controller **123** (and also controller **115**), though more or less sensors can be employed (alternatively, the present invention is not required to be used in conjunction with sensors, such as position sensors **535A**, **535B**, but the present invention can function by user controlling, for example, hydraulic remotes/valves manually, with sensors being absent). Each sensor **535A**, **535B** can be a position sensor for sensing a position

of windguard assembly **128**, so as to determine whether windguard assembly **128** is in first position **237**, second position **338**, or some intermediate position therebetween. Sensor **535A** can sense a position of actuator **251** (such as whether the rod extended or retracted), a position of arm **270** and/or shaft **275**, and/or an angular position of first linkage **276**. Sensor **535B** can be positioned to sense a position of third linkage **278** relative to fourth linkage **264**, a position of fourth linkage **264** and/or bar **259** relative to second section **246**, and/or a position of tines **260** relative to second section **246**. In sensing the position of third linkage **278** relative to fourth linkage **264**, sensor **535B** can be positioned on or near third or fourth linkage **278**, **264** and can be configured to sense an angular position of third linkage **278** relative to fourth linkage **264**. In sensing the position of fourth linkage **264** and/or bar **259** relative to second section **246**, sensor **535B** can sense an angular position of fourth linkage **264** and/or bar **259** relative to second section **246**. In sensing the position of tines **260** relative to second section **246**, sensor **535B** can sense a linear or angular position of tines **260** relative to second section **246**. Alternatively or in addition thereto, other position sensors can be used to sense the position of other structure that would be indicative of the position of windguard assembly **128**. Sensors **535A**, **535B** can output their position signals (corresponding to the sensed position of the referenced structures) to controller **123** (and/or controller **115**), which can receive the position signals and output an adjustment signal to feeder system **108** based upon these position signals. The adjustment signals can reposition windguard assembly **128** depending upon the circumstances, and/or change the direction of rotors **144**, in order to unplug crop material **136** or to resume normal operations.

(30) Alternatively or in addition thereto, control system **129** can be configured to automatically unplug feeder system **108** of a plug of crop material **136**, without the user having to enter a command to do so (this alternative can occur with a user operating tractor **100** pulling baler **101**, or with autonomous tractor and/or baler operations). Though not shown, control system **122** of baler **101** can include at least one sensor configured for: (a) sensing a plug of crop material **136** in feeder system **108** (for example, optical sensors detecting a profile of crop material **136** indicative of a plug of crop material, and/or weight sensors detecting a weight of crop material **136** indicative of a plug of crop material **136**, and/or any other sensor(s) suitable for detecting a plug of crop material **136**); and (b) outputting a plug signal corresponding to the plug sensed by the sensor. Controller **123** (and/or controller **115**) can be configured for: (a) receiving this plug signal; (b) for outputting an adjustment signal, based at least in part on the plug signal, so as to adjust windguard assembly **128** so as to remove the plug of crop material **136** and thereby to move windguard assembly to second position **338**; and (c) after removal of the plug, for outputting an additional adjustment signal, based at least in part upon another position signal indicative of windguard assembly **128** being in second position **338** and another signal indicative of the removal of the plug of crop material (i.e., another optical signal indicating a profile, or lack thereof, of crop material **136**, or a weight or mass of crop material **136**, indicating an absence of the plug), for moving windguard assembly **128** back to first position **237** for normal operation of baler **101**.

(31) In use, windguard assembly **128** can initially be in first position **237**, the normal operating position. To remove a crop plug, latch **271** can be pivoted counter-clockwise (either manually by a rope or cable connected to hole **381**, or by an actuator, or another trigger associated with baler **101**); for instance, control system **129** may be configured to prompt the user to release latch **271** during an automatic series of actions. Upon doing so, by action of crop material in reverse (such as by way of reversing rotors **144**, by way of reversing tines **240** of pickup assembly **111**, and/or by moving baler **101** backwards) arm **270** can be pushed forward by crop material and thereby slid from rearward position **272** to forward position **373** in slots **274**, which places windguard assembly **128** in second position **338**. This sliding action can be triggered by a command inputted into control system **129** by user to remove a crop plug in feeder system **108** (such as by reversing rotors **144**), or a crop plug can be sensed by control system **129**, which then can automatically cause a series of actions so as to automatically remove the crop plug (such as with, but not limited to,

automatic tractor and/or baler operations), though control system **129** may prompt user to release latch **271**. Alternatively, actuator **251** can move windguard assembly **121** forward, or a user can do this manually. Positioning windguard assembly **128** in second position **338** creates space and prevents tines **260** from acting as barbs relative to the crop plug when the crop flow is reversed (**133**), with arm **270** forward in forward position **373** and tine mechanism **248** pivoted to up position **368**. After removing the crop plug, windguard assembly **128** can be returned to first position **237**, by driving tractor **100** and thus baler **101** forward into the ejected pile of crop material **136** and/or the windrow of crop material **136**, so that the crop material **136** pushes windguard assembly **128** back into first position **237** (this can be assisted by rotating rotors **144** in their normal direction **134**). Alternatively, windguard assembly **128** can be returned to first position **237** manually, or automatically by control system **129**. When shaft **275** of arm **270** contacts a leading edge of latch **271** as arm **270** slides rearward in slots **274**, latch **271** can be caused to pivot up so that arm **270** can proceed rearwardly and seat in rearward position **272**, with tine mechanism **248** again in down position **267**. Further, a user may find it advantageous to move windguard assembly to second position **338** (or at some intermediate position between first and second positions **237**, **338**) in order to create space in, and thereby improve accessibility to, feeder system **108** and thereby to conduct maintenance of feeder system **108** or any other aspect of baler **101**, such as for maintenance of pickup assembly **111**, rotary feeder assembly **121**, windguard assembly **128**, and/or for replacement of tines **240** or hoops (i.e., which tines **240** pass between).

(32) Further, yet another embodiment of the present invention is provided. That is, the windguard assembly of the present invention can be mounted to baler tongue **107**. Thus, any of the embodiments of the windguard assembly according to the present invention can be adapted so as to mount to tongue **107**. Further, in accordance with yet another embodiment of the present invention, the windguard assembly may omit the roller, or may have a baffle in place of tines.

(33) Further, according to yet another embodiment of the present invention, the windguard assembly can move between the first and second positions as the windguard assembly rides atop the crop material when the rotor feed assembly **121** is reversed. The primary difference, however, is that the windguard assembly does not have a sliding action within a slot, such as slot **274**. Instead, the windguard assembly can primarily be a four-bar linkage, which rotates and which moves between a retracted position and an extended position. This embodiment is not specifically shown.

(34) Referring now to FIG. **6**, there is shown a flow diagram showing a method **690** of using an agricultural baler **101**, the method **690** including the steps of: providing **691** a baler frame **130** and a feeder system **108**, the feeder system **108** being coupled with the baler frame **130** and including a windguard assembly **128**; occupying **692** selectively, by the windguard assembly **128**, a first position **237**, the first position **237** being associated with a normal flow direction **132** of a crop material **136** relative to the feeder system **108**; and occupying **693** selectively, by the windguard assembly **128**, a second position **338** by moving between the first position **237** and the second position **338**, the second position **338** being associated with a reversal **133** of the normal flow direction **132** of the crop material **136**. The windguard assembly **128** can be configured for selectively occupying the first position **238** and the second position **339** by moving therebetween and thereby by sliding therebetween. The feeder system **108** includes a pickup assembly **111** and a rotary feeder assembly **121** each of which is coupled with the baler frame **130**, the windguard assembly **128** being coupled with the pickup assembly **111**, the rotary feeder assembly **121** including a rotor device **144**, the second position **338** being associated with a reversed operation of the rotor device **144**. The windguard assembly **128** can include an arm **270** slidably coupled with the pickup assembly **111** such that the arm **270** is configured for sliding between a rearward position **272** and a forward position **373**, the rearward position **272** being associated with the first position **237**, the forward position **373** being associated with the second position **338**. The windguard assembly **128** can include: a tine mechanism **248** pivotably coupled with the arm **270**; and a linkage assembly **250** coupled with the pickup assembly **111**, the arm **270**, and the tine

mechanism **248** and configured for moving the tine mechanism **248** between a down position **267** and an up position **368** when the arm **270** slides between the rearward position **272** and the forward position **373**, respectively. The linkage assembly **250** can be rigidly coupled with the tine mechanism **248**, the pickup assembly **111** further including a latch **271** configured for selectively latching the arm **270** in the first position **237**.

(35) It is to be understood that the steps of method **690** may be performed by controller **115**, **123** upon loading and executing software code or instructions which are tangibly stored on a tangible computer readable medium, such as on a magnetic medium, e.g., a computer hard drive, an optical medium, e.g., an optical disc, solid-state memory, e.g., flash memory, or other storage media known in the art. Thus, any of the functionality performed by controller **115**, **123** described herein, such as the method **690**, is implemented in software code or instructions which are tangibly stored on a tangible computer readable medium. The controller **115**, **123** loads the software code or instructions via a direct interface with the computer readable medium or via a wired and/or wireless network. Upon loading and executing such software code or instructions by controller **115**, **123**, controller **115**, **123** may perform any of the functionality of controller **115**, **123** described herein, including any steps of the method **690**.

(36) The term “software code” or “code” used herein refers to any instructions or set of instructions that influence the operation of a computer or controller. They may exist in a computer-executable form, such as machine code, which is the set of instructions and data directly executed by a computer's central processing unit or by a controller, a human-understandable form, such as source code, which may be compiled in order to be executed by a computer's central processing unit or by a controller, or an intermediate form, such as object code, which is produced by a compiler. As used herein, the term “software code” or “code” also includes any human-understandable computer instructions or set of instructions, e.g., a script, that may be executed on the fly with the aid of an interpreter executed by a computer's central processing unit or by a controller.

(37) These and other advantages of the present invention will be apparent to those skilled in the art from the foregoing specification. Accordingly, it is to be recognized by those skilled in the art that changes or modifications may be made to the above-described embodiments without departing from the broad inventive concepts of the invention. It is to be understood that this invention is not limited to the particular embodiments described herein, but is intended to include all changes and modifications that are within the scope and spirit of the invention.

Claims

1. A windguard assembly for a feeder system of an agricultural baler, the agricultural baler including a baler frame, the feeder system being coupled with the baler frame, the feeder system including a pickup assembly, the windguard assembly comprising: an arm configured to be pivotably coupled to the pickup assembly; a roller rotatably coupled to the arm, the roller having a first position with respect to the pickup assembly associated with a normal flow direction of a crop material relative to the feeder system and a second position with respect to the pickup assembly associated with a reversal of the normal flow direction of the crop material; a tine mechanism pivotably coupled with the arm; and a linkage assembly coupled with the pickup assembly, the arm, and the tine mechanism and configured for moving the tine mechanism between a down position when the roller is in the first position and an up position when the roller is in the second position.
2. The windguard assembly of claim 1, wherein the second position is associated with a reversed operation of a rotor device of a rotary feeder assembly, which is coupled with the baler frame.
3. The windguard assembly of claim 2, wherein the arm is further configured for being slidably coupled with the pickup assembly such that the arm is configured for sliding between a rearward position and a forward position, the rearward position being associated with the first position, the forward position being associated with the second position.

4. The windguard assembly of claim 1, wherein the linkage assembly is rigidly coupled with the tine mechanism.

5. An agricultural machine, comprising: a machine frame; and a feeder system coupled with the machine frame and including a pickup assembly and a windguard assembly, the windguard assembly comprising: an arm configured to be pivotably coupled to the pickup assembly; a roller rotatably coupled to the arm, the roller having a first position with respect to the pickup assembly associated with a normal flow direction of a crop material relative to the feeder system and a second position with respect to the pickup assembly associated with a reversal of the normal flow direction of the crop material; a tine mechanism pivotably coupled with the arm; and a linkage assembly coupled with the pickup assembly, the arm, and the tine mechanism and configured for moving the tine mechanism between a down position when the roller is in the first position and an up position when the roller is in the second position.

6. The agricultural machine of claim 5, wherein the agricultural machine is an agricultural baler and the machine frame is a baler frame, the feeder system including a rotary feeder assembly coupled with the baler frame, the rotary feeder assembly including a rotor device, the second position being associated with a reversed operation of the rotor device.

7. The agricultural machine of claim 6, wherein the arm is slidably coupled with the pickup assembly such that the arm is configured for sliding between a rearward position and a forward position, the rearward position being associated with the first position, the forward position being associated with the second position.

8. The agricultural machine of claim 5, wherein the linkage assembly is rigidly coupled with the tine mechanism, the pickup assembly further including a latch configured for selectively latching the arm in the first position.

9. A method of using an agricultural baler, the method comprising the steps of: providing a baler frame and a feeder system, the feeder system being coupled with the baler frame and including a pickup assembly and a windguard assembly, the windguard assembly comprising an arm configured to be pivotably coupled to the pickup assembly, a roller rotatably coupled to the arm, the roller having a first position with respect to the pickup assembly associated with a normal flow direction of a crop material relative to the feeder system and a second position with respect to the pickup assembly associated with a reversal of the normal flow direction of the crop material, a tine mechanism pivotably coupled with the arm, and a linkage assembly coupled with the pickup assembly, the arm, and the tine mechanism and configured for moving the tine mechanism between a down position when the roller is in the first position and an up position when the roller is in the second position; occupying selectively, by the roller, a first position, with the tine mechanism in the down position, the first position being associated with a normal flow direction of a crop material relative to the feeder system; and occupying selectively, by the roller, a second position with the tine mechanism in the down position the second position being associated with a reversal of the normal flow direction of the crop material.

10. The method of claim 9, wherein the windguard assembly is configured for selectively occupying the first position and the second position by moving therebetween and thereby by sliding therebetween, the feeder system including a rotary feeder assembly coupled with the baler frame, the rotary feeder assembly including a rotor device, the second position being associated with a reversed operation of the rotor device.

11. The method of claim 10, wherein the arm is configured for sliding between a rearward position and a forward position, the rearward position being associated with the first position, the forward position being associated with the second position.

12. The method of claim 9, wherein the linkage assembly is rigidly coupled with the tine mechanism, the pickup assembly further including a latch configured for selectively latching the arm in the first position.
